Experiment ‘p_6_2_i_has_adjacent_duplicates_seq’ Results

June 5, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` : 0
Responses containing `expect` : 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_6_2_i_has_adjacent_duplicates_seq
Natural language statement: null
Method signature: p_6_2_i_has_adjacent_duplicates_seq(inputs: seq<int>) returns (has_duplicates: bool)

Ensures

- `has_duplicates == (exists i :: 0 <= i < |inputs| - 1 && inputs[i] == inputs[i + 1])`

Requires

- `|inputs| >= 2`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

p_6_2_i_has_adjacent_duplicates_seq(inputs: seq<int>) returns (has_duplicates: bool)

The method should respect the following contract:

requires |inputs| >= 2, ensures has_duplicates == (exists i :: 0 <= i < |inputs| - 1 &&
    ↪ inputs[i] == inputs[i + 1])

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DnO6U7ZSkXOzqE5ByxLrBtIyI8xVZ
Model: gpt-5.5-2026-04-23
Created at: 1780663098
Finish reason: stop
Usage: 316 tokens in, and 4538 tokens out

```
{
  var i := 0;
  while i < |inputs| - 1
    invariant 0 <= i <= |inputs| - 1
    invariant forall j :: 0 <= j < i ==> inputs[j] != inputs[j + 1]
    decreases |inputs| - 1 - i
  {
    if inputs[i] == inputs[i + 1] {
      has_duplicates := true;
      assert 0 <= i < |inputs| - 1 && inputs[i] == inputs[i + 1];
      assert exists j :: 0 <= j < |inputs| - 1 && inputs[j] == inputs[j + 1];
      return;
    }

    assert inputs[i] != inputs[i + 1];
    var oldI := i;
    forall j | 0 <= j < oldI + 1
      ensures inputs[j] != inputs[j + 1]
    {
      if j < oldI {
```

```
        assert inputs[j] != inputs[j + 1];
      } else {
        assert j == oldI;
        assert inputs[j] != inputs[j + 1];
      }
    }
    i := i + 1;
  }

  assert i == |inputs| - 1;
  assert !(exists j :: 0 <= j < |inputs| - 1 && inputs[j] == inputs[j + 1]) by {
    if (exists k :: 0 <= k < |inputs| - 1 && inputs[k] == inputs[k + 1]) {
      var j :| 0 <= j < |inputs| - 1 && inputs[j] == inputs[j + 1];
      assert inputs[j] != inputs[j + 1];
      assert false;
    }
  }
  has_duplicates := false;
}
```

Final Program

```
method p_6_2_i_has_adjacent_duplicates_seq(inputs: seq<int>) returns (has_duplicates:
    ↪ bool)
        requires |inputs| >= 2
        ensures has_duplicates == (exists i :: 0 <= i < |inputs| - 1 && inputs[i] ==
            ↪ inputs[i + 1])
{
  var i := 0;
  while i < |inputs| - 1
    invariant 0 <= i <= |inputs| - 1
    invariant forall j :: 0 <= j < i ==> inputs[j] != inputs[j + 1]
    decreases |inputs| - 1 - i
  {
    if inputs[i] == inputs[i + 1] {
      has_duplicates := true;
      assert 0 <= i < |inputs| - 1 && inputs[i] == inputs[i + 1];
      assert exists j :: 0 <= j < |inputs| - 1 && inputs[j] == inputs[j + 1];
      return;
    }

    assert inputs[i] != inputs[i + 1];
    var oldI := i;
    forall j | 0 <= j < oldI + 1
      ensures inputs[j] != inputs[j + 1]
    {
      if j < oldI {
        assert inputs[j] != inputs[j + 1];
      } else {
        assert j == oldI;
```

```
      assert inputs[j] != inputs[j + 1];
    }
  }
  i := i + 1;
}

assert i == |inputs| - 1;
assert !(exists j :: 0 <= j < |inputs| - 1 && inputs[j] == inputs[j + 1]) by {
  if (exists k :: 0 <= k < |inputs| - 1 && inputs[k] == inputs[k + 1]) {
    var j :| 0 <= j < |inputs| - 1 && inputs[j] == inputs[j + 1];
    assert inputs[j] != inputs[j + 1];
    assert false;
  }
}
has_duplicates := false;
}
```

Total Token Usage

Input tokens: 316
Output tokens: 4538
Reasoning tokens: 4066
Sum of 'total tokens': 4854

Experiment Timings

Overall Experiment started at 1780663097095, ended at 1780663207652, lasting 110557ms (110.56 seconds)
Iteration #1 started at 1780663097095, ended at 1780663207652, lasting 110557ms (110.56 seconds)